

A Computationally-Integrated SF-PBL Framework for Engineering Dynamics: Design, Implementation, and Mixed-Methods Evaluation in an OBE-Aligned Undergraduate Course

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Abstract

Engineering dynamics courses demand simultaneous development of mathematical modelling, computational implementation, and communication competencies, yet conventional instruction rarely provides authentic contexts for their integration. This study presents the design, implementation, and mixed-methods evaluation

of a Simulation-Focused Problem-Based Learning (SF-PBL) framework embedded within a 15-week undergraduate Dynamic System Modeling (DSM) course ($N = 53$, B.Tech Mechatronics Engineering, Semester IV, January–April 2026) at an NBA-accredited institution. The framework operationalises a four-phase pipeline — System Identification, Mathematical Modelling, Computational Simulation, Validation & Communication — scaffolded through an Explain-Try-Challenge model and assessed via a six-judge expert panel. A pre–post mixed-methods design measured self-efficacy (10-item Likert scale), conceptual knowledge (5-item MCQ), PBL satisfaction, AI tool usage, and qualitative reflections. Self-efficacy improved significantly (pre: $M = 2.92$, $SD = 0.21$; post: $M = 3.80$, $SD = 0.40$; $t(52) = 13.91$, $p < .001$, Cohen’s $d = 2.72$, pooled SD). Knowledge gains were large ($t(52) = 11.32$, $p < .001$, $d = 1.20$; Hake’s $g = 0.50$, medium band). All ten self-efficacy items improved significantly ($p < .001$). PBL satisfaction was high ($M = 3.74$, $SD = 0.30$). Eighty-five percent used AI tools, primarily for computational coding. Qualitative themes identified simulation validation, PID tuning, and linking derivation to computation as breakthrough moments. Exhibition rubric scores from 17 student projects ($M = 82.62/100$, $SD = 9.98$) confirmed strong performance across all domains. The SF-PBL framework provides a replicable, evidence-based model for computationally-integrated dynamics pedagogy in OBE-aligned STEM programmes.

Keywords: problem-based learning; engineering dynamics; computational simulation; self-efficacy; OBE; authentic assessment; AI-assisted learning; STEM education

1 Introduction

1.1 The Pedagogical Challenge in Engineering Dynamics

Engineering dynamics — encompassing kinematics, constitutive laws, Lagrangian mechanics, and vibration analysis — occupies a foundational position in mechanical and mechatronics engineering curricula. It presents persistent pedagogical challenges: the mathematical formalism is demanding, the physical intuition difficult to develop, and the connection to real engineering systems rarely made explicit within traditional lecture-driven courses (Streveler et al., 2008). Students frequently solve textbook problems through surface-level pattern matching without building the conceptual understanding required for engineering practice.

Compounding this challenge is the increasing centrality of computational tools. Modern engineers are expected not only to derive equations of motion analytically but to implement them numerically, simulate dynamic system behaviour, and interpret computational outputs — a set of competencies that conventional closed-book assessments rarely evaluate authentically (Justo et al., 2021). The emergence of AI coding assistants (ChatGPT, GitHub Copilot, Claude) has further transformed the competency landscape: students can now generate simulation code rapidly, raising important questions about depth of conceptual understanding versus tool-assisted productivity.

1.2 Problem-Based Learning as a Solution Pathway

Problem-Based Learning (PBL), as defined by Hmelo-Silver (2004), places authentic, ill-structured problems at the centre of the learning experience, requiring students to self-direct their inquiry, build domain knowledge, and communicate solutions collaboratively. In engineering education, PBL has been shown to improve conceptual understanding, motivation, and collaborative competencies (Prince and Felder, 2006; Dym et al., 2005). However, its implementation in mathematically intensive courses such as dynamics remains underexplored, particularly when computational simulation is foregrounded as the primary learning artefact.

1.3 The Research Gap

Three specific gaps motivate this work:

1. **Computational-PBL integration gap:** Existing PBL implementations in engineering dynamics rarely position Python-based ODE simulation as the core learning product, leaving the derivation-to-computation bridge unaddressed.
2. **Authentic assessment gap:** Multi-expert external evaluation panels covering industry, intellectual property, mathematics, and communication are uncommon in undergraduate PBL contexts; most studies rely on instructor-only assessment.
3. **OBE alignment gap:** Few published studies demonstrate explicit, auditable traceability from Course Outcomes (COs) through Bloom’s taxonomy levels to rubric dimensions within an accreditation-compliant (NBA/ABET) framework.

1.4 Purpose and Contributions

This paper presents the design, implementation, and empirical evaluation of the SF-PBL framework embedded within the Dynamic System Modeling (DSM) course (702MH0C023), a 15-week core course for B.Tech Mechatronics Engineering students at NMIMS MPSTME, Mumbai (2 January – 25 April 2026). The study makes four contributions:

1. A documented, replicable **four-phase SF-PBL computational algorithm** (System Identification → EOM Derivation → Python Simulation → Validation & Communication)
2. An **OBE-aligned rubric framework** with complete CO→Bloom’s→NBA Programme Outcome traceability
3. **Empirical evidence** of large-to-very-large learning gains (Cohen’s $d > 1.0$ on both self-efficacy and knowledge) from a mixed-methods pre–post study ($N = 53$)
4. **Authentic assessment data** from a six-judge expert panel evaluating 17 student projects across five engineering domains

2 Theoretical Framework

2.1 Simulation-Focused Problem-Based Learning (SF-PBL)

SF-PBL is defined in this study as a variant of PBL in which **mathematical derivation from first principles and its subsequent computational simulation constitute the primary learning artefact**. This distinguishes SF-PBL from project-based learning (where the product is typically a physical prototype) and from standalone computational modelling courses (where system identification and mathematical modelling steps are usually pre-specified).

The SF-PBL framework rests on three theoretical pillars. First, **Hmelo-Silver's PBL cycle** (2004): students encounter an ill-structured real engineering system, engage in self-directed inquiry to identify governing physics, construct knowledge through mathematical and computational means, and reflect collaboratively on outcomes. Second, **Bandura's self-efficacy theory** (1997): mastery experiences — successfully deriving equations of motion and validating them in simulation — are the most potent source of self-efficacy gains. The sequential phases of SF-PBL are designed to provide escalating mastery experiences mapped directly to course competencies. Third, **Hake's interactive engagement framework** (1998): courses achieving normalised gain $g > 0.30$ are classified as effectively interactive-engagement; SF-PBL's four-phase structure, with active derivation, implementation, and public defence, is hypothesised to achieve medium-to-high normalised gain.

2.2 OBE Framework and NBA Alignment

The DSM course was designed using Outcome-Based Education (OBE) principles (Spady, 1994) with explicit backward design from Programme Outcomes (POs) to Course Outcomes (COs) to assessment rubric dimensions. Five Course Outcomes (CO1–CO5) are detailed in Table 1.

Table 1: Course Outcomes and Bloom’s Taxonomy Levels

CO	Statement	Bloom’s Level
CO1	Apply kinematic analysis to multi-body systems	Apply (L3)
CO2	Construct FBDs for complex mechanisms	Analyse (L4)
CO3	Derive EOMs using Newton’s laws	Analyse (L4)
CO4	Apply Lagrangian mechanics to holonomic systems	Evaluate (L5)
CO5	Implement ODE solvers in Python and validate	Create (L6)

2.3 The Explain-Try-Challenge (E-T-C) Instructional Model

Each 90-minute lecture session followed a three-phase model: **Explain** (30 min) — conceptual introduction with worked examples; **Try** (40 min) — guided practice on structured problems using Jupyter notebooks; **Challenge** (20 min) — open-ended problem connecting lecture content to the student’s own SF-PBL project system.

3 Methodology

3.1 Research Design

This study employed a **single-cohort pre-post mixed-methods design** (Creswell and Creswell, 2018) comprising:

- *Quantitative strand:* Pre-post surveys (10-item self-efficacy Likert scale, 5-item knowledge MCQ, 10-item PBL satisfaction scale), categorical AI usage data, and six-judge exhibition rubric scores
- *Qualitative strand:* Thematically coded open-ended reflections (OE1–OE3) from the post-survey
- *Triangulation:* Quantitative learning gains corroborated by qualitative breakthrough narratives referencing specific SF-PBL phases

3.2 Course Context and Participants

Course parameters and participant details are presented in Table 2.

Table 2: Course Context and Participant Summary

Parameter	Value
Course code	702MH0C023 — Dynamic System Modeling
Institution	NMIMS MPSTME, Mumbai (NBA-accredited)
Programme	B.Tech Mechatronics Engineering, Sem. IV
Academic year	2025–26
Course dates	2 January 2026 – 25 April 2026
Duration	15 working weeks (3 credits: 2L + 2P/week)
N registered	55 students
N consenting & participating	53 (2 did not provide informed consent)
N present at exhibition	53 (all consenting participants)
Teams	17 (2–4 members per team)
Engineering project domains	5

Participation in the research survey was voluntary and anonymised; course grades were not affected by research participation. Informed consent was obtained from all 53 participants prior to pre-survey administration. Two registered students did not provide consent and are excluded from all analyses ($N = 53$ throughout).

3.3 The SF-PBL Framework: Four-Phase Computational Algorithm

The SF-PBL project ran concurrently with the 15-week course (2 January – 25 April 2026) and culminated in a public Mini Project Exhibition on **25 April 2026** at NMIMS MPSTME, organised in collaboration with the NMIMS IUCEE Student Chapter (video documentation available via LinkedIn: <https://www.linkedin.com/posts/nmims-iucee-student-chapter-miniprojectexhibition2026-sfpbl-simulationfocusedpbl-activity-7455575150122471424-nQ9>). Each of the 17 teams executed the following four-phase pipeline:

Phase 1: System Identification (Weeks 1–4). Students selected a real engineering system (student-chosen, not prescribed), identified degrees of freedom (DOF) and constraints, and defined coordinate frames. *Deliverable:* System description + kinematic sketch. CO1 | Bloom’s Apply (L3).

Phase 2: Mathematical Modelling (Weeks 5–9). Students drew Free Body Diagrams (FBDs) by hand, applied Newton’s Second Law or the Lagrangian formulation, derived coupled Equations of Motion (EOMs), and converted to state-space form $\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u})$. *Deliverable:* Analytical EOMs + state-space representation. CO2, CO3, CO4 | Bloom’s Analyse–Evaluate (L4–L5).

Phase 3: Computational Simulation (Weeks 10–13). Students implemented the ODE solver in Python using `scipy.integrate.solve_ivp`, generated time-domain plots (position, velocity, energy vs. time), and tuned control parameters (e.g., PID gains). *Deliverable:* Working simulation + validated plots. CO5 | Bloom’s Create (L6).

Phase 4: Validation & Communication (Weeks 14–15). Students validated simulations against energy conservation, limit cases, or published benchmarks, then presented to the six-judge expert panel at the public exhibition. *Deliverable:* Exhibition presentation + validation evidence. CO5 | Bloom’s Evaluate–Create (L5–L6).

3.4 The Six-Judge Expert Evaluation Panel

To ensure authentic, multi-perspective, and bias-reduced assessment, a six-member external expert panel was constituted spanning five professional domains (Table 3).

Table 3: Six-Judge Expert Evaluation Panel (25 April 2026)

#	Judge	Affiliation	Domain	Max
1	Kamlesh Panchal	Manufacturing Industry	Production Engg.	/16
2	Hasti Chandarana	IP Consultancy	Patent & Innovation	/16
3	Nitu Gupta	External Academic	Mathematics	/16
4	Mahendra Kane (Siemens)	Siemens Ltd.	Industry 4.0 / DT	/20
5	Parminder Jandoo	Training Organisation	Communication	/16
6	Debasis Dash	HR Consultancy	Behavioural Science	/16
TOTAL				/100

3.5 Student Projects: 17 Teams, 5 Engineering Domains

The 17 student teams were distributed across five project categories: (A) Robotics & Locomotion — 7 teams (humanoid, hexapod, multi-terrain robot, quadruped, pick-and-

place rover, robotic gripper, H₂ fuel cell UAV); (B) Aerospace & Marine — 3 teams (rocket landing & docking, ROV, fixed-wing UAV); (C) Control Systems — 2 teams (ball balancing Stewart platform, self-balancing robot); (D) Vehicle Dynamics — 2 teams (critical skid velocity, shock absorber); (E) Electromechanical — 3 teams (digital twin DC motors, motor modelling, automated EoT crane).

3.6 Survey Instruments

Self-Efficacy Scale (SE1–SE10): Ten items rated on a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree), each aligned to a specific CO, covering kinematic analysis, FBD construction, Newton’s EOM derivation, Lagrangian formulation, state-space conversion, Python ODE implementation, energy validation, PID tuning, plot interpretation, and technical communication.

Knowledge MCQ (PK1–PK5): Five items assessing degrees of freedom, Lagrangian formulation, pendulum natural frequency, viscous damping effects, and Newton’s rotation law. Scored 0–5.

PBL Experience Scale (PBL1–PBL10): Ten items on a 5-point scale, grouped into perceived value (PBL1–3), scaffold quality (PBL4–7), and overall satisfaction (PBL8–10).

AI Tool Usage: Multi-select categorical (ChatGPT, GitHub Copilot, Claude, None, Other); phase of primary use; estimated percentage of mathematical derivation completed without AI.

Open-Ended Reflections (OE1–OE3): (OE1) Most valuable learning; (OE2) Suggested changes for future cohorts; (OE3) Breakthrough moment when simulation made sense.

3.7 Data Analysis

- **Paired *t*-tests** (two-tailed, $\alpha = .05$): Pre-post SE and PK comparisons
- **Cohen’s *d*** (pooled SD: $d = \Delta M / \sqrt{(SD_1^2 + SD_2^2)/2}$): small = 0.2, medium = 0.5, large = 0.8 (Cohen, 1988)

- **Hake’s normalised gain:** $g = (\text{post} - \text{pre}) / (\text{max} - \text{pre})$ per student; bands: low < 0.30 , medium $0.30\text{--}0.69$, high ≥ 0.70 (Hake, 1998)
- **95% CI:** $\pm 1.96 \times SE_{\text{mean}}$
- **Wilcoxon signed-rank test:** non-parametric confirmation ($\alpha = .05$)
- **Post-hoc power analysis** (G*Power framework)
- **Thematic analysis** (Thomas and Harden, 2008): OE1–OE3 coded by frequency of theme occurrence

All analyses were conducted in Python 3.10 using SciPy 1.11 and NumPy 1.25. Analysis scripts are available from the corresponding author on request.

4 Results

4.1 Self-Efficacy: Pre–Post Comparison

Self-efficacy in dynamic system modelling increased significantly from pre- to post-exhibition (Table 4). The mean gain of +0.88 points (Cohen’s $d = 2.72$) represents a very large effect substantially exceeding the threshold for practical significance (Cohen, 1988). The result was confirmed by the non-parametric Wilcoxon signed-rank test ($W = 0.0$, $p < .001$) with post-hoc power = 1.000.

Table 4: Self-Efficacy Pre–Post Comparison ($N = 53$)

Measure	Pre M (SD)	Post M (SD)	Gain	95% CI	$t(52)$	p	d
SE Mean (1–5)	2.92 (0.21)	3.80 (0.40)	+0.88	[0.75, 1.00]	13.91	<.001	2.72

All ten individual SE items were statistically significant at $p < .001$ (Table 5). The largest gains were observed for SE8 (PID Tuning, $d = 2.22$) and SE3 (Newton’s EOM, $d = 2.14$), both directly activated in Phases 2 and 3. The smallest gain, SE5 (State-Space, $d = 1.46$), remains a very large effect and reflects the advanced junction between analytical derivation and computational implementation.

Table 5: Per-Item Self-Efficacy Results ($N = 53$). All $p < .001$.

Item	Competency Focus	Pre M	Post M	Gain	d
SE1	Kinematics	2.89	3.79	+0.91	1.69
SE2	Free Body Diagram	2.94	3.77	+0.83	1.72
SE3	Newton's EOM	2.89	3.81	+0.92	2.14
SE4	Lagrangian	2.93	3.81	+0.89	1.85
SE5	State-Space	2.93	3.72	+0.79	1.46
SE6	ODE / Python	2.93	3.91	+0.98	1.82
SE7	Energy Validation	3.02	3.77	+0.76	1.63
SE8	PID Tuning	2.91	3.83	+0.92	2.22
SE9	Plot Interpretation	2.89	3.77	+0.89	1.69
SE10	Technical Communication	2.91	3.77	+0.87	1.57

Cohen's d range: 1.46 (SE5) – 2.22 (SE8). All d values large by Cohen (1988).

4.2 Conceptual Knowledge: Pre–Post Comparison

Knowledge MCQ scores increased significantly (Table 6). Hake's normalised gain $g = 0.50$ falls in the **medium** interactive-engagement category (Hake, 1998) ($0.30 \leq g < 0.70$), in the upper half of the medium band. The MCQ distribution shift — from left-skewed (pre, mode = 2, no student achieving 5/5) to right-skewed (post, 12 students achieving 5/5) — confirms broad-based conceptual development (Table 7).

Table 6: Knowledge MCQ Pre–Post Comparison ($N = 53$)

Measure	Pre M (SD)	Post M (SD)	Gain	95% CI	$t(52)$	p	d	g
PK Total (0–5)	1.96 (0.94)	3.28 (1.25)	+1.32	[1.09, 1.55]	11.32	<.001	1.20	0.50

Table 7: MCQ Score Distribution ($N = 53$)

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$
Pre (n)	3	11	28	7	4	0
Post (n)	0	4	11	16	10	12

4.3 PBL Experience Satisfaction

Post-exhibition PBL satisfaction was consistently high across all three sub-scales (Table 8), with a narrow SD of 0.30 indicating high cohort-level consistency in the SF-PBL experience.

Table 8: PBL Experience Scale — Post-Survey ($N = 53$)

Sub-scale	Items	M	SD
PBL Value	PBL1–3	3.70	—
Scaffold Quality	PBL4–7	3.76	—
Overall Satisfaction	PBL8–10	3.76	—
Total Mean	PBL1–10	3.74	0.30

4.4 Exhibition Rubric Scores: Six-Judge Panel

The combined six-judge assessment of 17 teams ($N = 53$ present students) yielded $M = 82.62/100$, $SD = 9.98$, range = 53–96 (Table 9).¹ The SD of approximately 10 points reflects meaningful discriminative ability while confirming that all present students achieved a substantive standard.

Table 9: Six-Judge Panel Scores ($N = 53$ students, 17 teams, 25 April 2026)

Judge	Domain	Max	M	SD	Range
Kamlesh Panchal	Manufacturing	/16	13.94	1.23	11–16
Hasti Chandarana	IP / Patent	/16	12.53	2.68	8–16
Nitu Gupta	Mathematics (External)	/16	12.77	2.67	6–16
Mahendra Kane (Siemens)	Industry 4.0	/20	17.09	2.18	12–20
Parminder Jandoo	Communication	/16	13.26	2.13	5–16
Debasis Dash	Behavioural Science	/16	13.02	2.28	8–16
COMBINED TOTAL		/100	82.62	9.98	53–96

4.5 AI Tool Usage

Forty-five of 53 students (84.9%) reported using AI tools during the SF-PBL project (Table 10). The most common use case was Python simulation coding (Phase 3) and debugging; 39.6% estimated 50–75% of their mathematical derivation was completed without AI assistance, indicating human-led mathematical reasoning remained dominant throughout.

¹One registered student was absent on exhibition day and scored 0/100; this student is excluded from all combined statistics. Including the absent student: $M = 81.09$, $SD = 14.97$, $N = 54$.

Table 10: AI Tool Usage Distribution ($N = 53$)

Tool / Combination	n	%
ChatGPT	18	34.0
ChatGPT + GitHub Copilot	11	20.8
None	8	15.1
Claude	6	11.3
ChatGPT + Claude	5	9.4
Other	5	9.4
Total AI users	45	84.9

4.6 Qualitative Themes

Thematic coding of open-ended post-survey responses revealed consistent patterns across all three open-ended items ($N = 53$):

OE1 — Most Valuable Learning: Mathematical validation of simulation results ($n = 18$, 34.0%); team collaboration ($n = 10$, 18.9%); debugging complex models ($n = 8$, 15.1%); other ($n = 17$, 32.1%).

OE3 — Breakthrough Moment: Linking mathematical derivation to simulation output ($n = 20$, 37.7%; e.g., “*The moment my simulation plot matched my analytical solution, I finally understood what the equations meant physically*”); understanding PID tuning from first principles ($n = 12$, 22.6%); seeing the physical model work in simulation ($n = 11$, 20.8%); other ($n = 10$, 18.9%).

OE2 — Suggested Changes: Smaller group sizes of 2–3 members ($n = 16$, 30.2%); less documentation, more coding time ($n = 12$, 22.6%); more project development time ($n = 11$, 20.8%); other ($n = 14$, 26.4%).

5 Discussion

5.1 Interpreting the Self-Efficacy Effect Size

The self-efficacy gains ($d = 2.72$) substantially exceed those reported in comparable PBL studies. Freeman et al. (2014)’s meta-analysis of active learning found a mean $d \approx 0.47$;

Dochy et al. (2003)’s PBL meta-analysis reported $d \approx 0.65$. The very large Cohen’s d here warrants careful interpretation: the narrow pre-survey SD (0.21 on a 5-point scale) reflects a relatively homogeneous cohort at baseline, which mathematically amplifies the standardised effect size. The absolute gain (+0.88 points, 95% CI [0.75, 1.00]) is arguably the more practically meaningful metric, representing a reliable cohort-wide shift from approximately “Neutral” to “Agree” across all ten competency items.

Critically, the two items with the largest gains — SE8 (PID Tuning, $d = 2.22$) and SE3 (Newton’s EOM, $d = 2.14$) — correspond directly to Phases 2 and 3 of the SF-PBL pipeline, confirming that the computational integration mechanism activated the intended learning outcomes rather than producing uniform, undifferentiated gains.

5.2 Knowledge Gains and Hake’s Normalised Gain

Hake’s normalised gain $g = 0.50$ places this course in the **medium** interactive-engagement category (Hake, 1998) ($0.30 \leq g < 0.70$), in the upper half of the medium band. This result is particularly notable given that: (a) the MCQ measured conceptual understanding rather than procedural recall; (b) students began with low prior knowledge (pre: $M = 1.96/5$, 39% correct); and (c) assessment under examination conditions reduced social desirability effects.

5.3 AI Integration: Complementarity Not Substitution

The finding that 84.9% of students used AI tools — primarily for Python simulation coding and debugging — suggests an emerging pattern of **human–AI complementarity** in SF-PBL. The framework’s structural design encouraged this: the rubric explicitly required hand-drawn FBDs and analytical EOM derivations *before* simulation (Phase 3), creating a curricular incentive to use AI as an accelerator for implementation rather than a substitute for mathematical reasoning. This contributes to the emerging discourse on AI in STEM education (Tlili et al., 2023): SF-PBL channels AI use productively by positioning derivation as the primary intellectual activity.

5.4 The Six-Judge Panel as Authentic Assessment

The six-judge panel spanning five professional domains represents a significant departure from conventional faculty-only assessment. The panel mean of 82.62/100 ($SD = 9.98$, range = 53–96) demonstrates that all present students achieved substantive scores with meaningful spread across team quality levels, consistent with Gulikers et al.’s (2004) criteria for authentic assessment validity. The Industry 4.0 judge (Siemens: $M = 17.09/20 = 85.5\%$) awarded the highest relative scores, suggesting student projects met contemporary digital engineering standards relevant for mechatronics graduates entering Industry 4.0 workplaces.

5.5 The Breakthrough Theme: Derivation-to-Simulation

The dominant qualitative breakthrough theme — *linking mathematical derivation to simulation output* ($n = 20$, 37.7%) — directly validates the theoretical premise of SF-PBL: that computation serves as the bridge between abstract mathematical formalism and physical intuition. This theme appeared independently of AI tool usage pattern, project complexity level, and team size, suggesting that the derivation-to-simulation moment is a **universal learning landmark** in computationally-integrated dynamics education requiring specific instructional scaffolding.

5.6 Practical Implications for STEM Educators

The SF-PBL framework offers three directly replicable elements: (1) the **four-phase pipeline** is domain-agnostic and transferable to thermodynamics, fluid mechanics, structural analysis, or control systems with minor adaptation; (2) the **E-T-C lecture model** provides an active-learning structure within existing contact hours without requiring curriculum restructuring; (3) the **six-judge rubric with CO→PO traceability** provides an accreditation-compliant (NBA/ABET) authentic assessment instrument with inter-professional credibility that faculty-only panels cannot achieve.

5.7 Limitations

Several limitations should be acknowledged. First, the study employed a single-group pre-post design without a concurrent control condition; causal attribution to the SF-PBL framework versus general course maturation effects is therefore constrained. Second, the sample ($N = 53$) is from a single institution, programme, and cohort, limiting generalisability. Third, AI usage data is self-reported and subject to social desirability bias. Fourth, formal inter-rater reliability (Cohen's κ) between the six expert judges was not computed, representing a measurement limitation for the rubric data. Fifth, the narrow pre-survey SD suggests possible range restriction in the self-efficacy baseline. Future research should address these limitations through controlled comparisons, multi-institution replication, longitudinal follow-up, and formal inter-rater reliability analysis for expert panels.

6 Conclusion

This study presented the design, implementation, and mixed-methods evaluation of the SF-PBL framework in an undergraduate engineering dynamics course at an NBA-accredited institution. The framework operationalises a four-phase computational pipeline — System Identification, Mathematical Modelling, Python Simulation, Validation & Communication — within a 15-week OBE-aligned course assessed by a six-judge external expert panel.

Principal findings:

- **Very large self-efficacy gains** across all ten competency items ($d = 2.72$, $p < .001$), largest for PID Tuning ($d = 2.22$) and Newton's EOM ($d = 2.14$)
- **Large knowledge gains** with medium Hake normalised gain ($g = 0.50$, upper half of medium band; $d = 1.20$, $p < .001$)
- **High PBL satisfaction** ($M = 3.74/5$, $SD = 0.30$) with high cohort consistency
- **Strong exhibition performance** ($M = 82.62/100$, $SD = 9.98$) across 17 project teams in five engineering domains
- **Emergent human-AI complementarity**: 84.9% AI usage concentrated in simulation coding; mathematical derivation predominantly human-led

- **Universal breakthrough moment:** Linking analytical EOM derivation to simulation output — the most frequently reported transformative learning experience (37.7%)

The SF-PBL framework provides a replicable, evidence-based, and OBE-compliant model for STEM educators seeking to integrate computational simulation authentically into engineering dynamics pedagogy. All instruments, rubrics, and analysis scripts are available from the corresponding author upon reasonable request.

Data Availability Statement

The anonymised individual-level survey dataset ($N = 53$, 65 variables), exhibition rubric evaluation scores (six judges across 17 teams), survey instruments, rubric specifications, Python analysis scripts (SciPy 1.11, NumPy 1.25), and representative student project artefacts are publicly available in the official repository: <https://github.com/sunny-nanade/SF-PBL-Engineering-Dynamics-2026>.

Ethics Statement

Informed consent was obtained from all participants prior to pre-survey administration. Participation was voluntary; course grades were not affected by non-participation. Data were anonymised for analysis. The study was conducted in accordance with institutional guidelines for educational research at NMIMS MPSTME.

Author Contributions

SN: Conceptualisation, course design, SF-PBL framework development, data collection, formal analysis, methodology, writing (original draft, review & editing), project administration.

Conflict of Interest

The author declares no conflict of interest.

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A Survey Instruments

A.1 Self-Efficacy Items (SE1–SE10)

Instructions: Rate your confidence in each ability on a 5-point scale (1 = Strongly Disagree, 5 = Strongly Agree).

SE1. I can apply kinematic analysis to identify DOF in a multi-body system. [CO1]

SE2. I can construct a correct Free Body Diagram for a complex mechanism. [CO2]

SE3. I can derive equations of motion using Newton’s Second Law. [CO3]

SE4. I can formulate the Lagrangian for a conservative dynamic system. [CO4]

SE5. I can convert a second-order ODE system to state-space form. [CO5]

SE6. I can implement a numerical ODE solver (`solve_ivp`) in Python. [CO5]

SE7. I can validate my simulation using energy conservation checks. [CO5]

SE8. I can tune PID controller gains to achieve a desired dynamic response. [CO5]

SE9. I can interpret simulation time-domain plots of position, velocity, and energy. [CO5]

SE10. I can communicate my simulation results clearly to a non-specialist. [Communication]

A.2 Knowledge MCQ Items (PK1–PK5)

Instructions: Select the single best answer.

PK1. A rigid body constrained to rotate in a plane has: (a) 3 DOF (b) **1 DOF** (c) 2 DOF (d) 0 DOF

PK2. The Lagrangian L is defined as: (a) $L = V - T$ (b) **$L = T - V$** (c) $L = T + V$ (d) $L = \partial T / \partial q$

PK3. The natural frequency of a simple pendulum (length l): (a) **$\omega_n = \sqrt{g/l}$** (b) $\omega_n = \sqrt{l/g}$ (c) $\omega_n = g/l$ (d) $\omega_n = l/g$

- 421 PK4. Adding viscous damping to an oscillator: (a) Increases natural frequency (b) Has
422 no effect (c) **Decreases the oscillation frequency** (d) Doubles the amplitude
- 423 PK5. Newton's Second Law for rotation states: (a) $\tau = ma$ (b) $\tau = I\alpha$ (c) $\tau = I\omega$
424 (d) $\tau = m\alpha$